Among all possible spanning trees, a *minimum spanning tree* (*MST*) is the one for which the sum of edges' weights is minimal.

Kruskal algorithm is beyond the scope of this book. Here it suffices to say that it constructs the MST for a graph by:

1. Starting with each vertex in a difference set;
2. Keeping a disjoint set of the graph vertices;
3. Going through the graph's edges **in order of increasing weight**;
4. For each edge, if its extremes are not in the same partition, merge their components;
5. If all vertices belong to the same component, stop.

The MST will be defined by the list of edges that triggers, at point (4), merge calls for the disjoint-set.

5.7.3 Clustering

Clustering is the most used unsupervised learning[15] algorithm. The problem here is that we would like to get a partitioning of a set of points into a few, usually disjoint, subsets.

There are several types of clustering algorithms. Although a taxonomy of clustering is beyond the scope of this chapter (we will devote chapter 12 to this topic), here we will mention one particular class of these algorithms.